Exploratory Data Analysis (EDA) Summary Report

1. Introduction

This report presents an exploratory data analysis (EDA) of Geldium's customer dataset, prepared for Tata iQ's analytics team and Geldium's decision-makers. The goals are to (1) assess the completeness, accuracy and consistency of the data, (2) treat missing and inconsistent values so the data is model-ready, and (3) identify early risk indicators that can guide the delinquency prediction model and Geldium's intervention strategy.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- Number of records: 500 customers (one row per customer) and 19 columns. No duplicate rows and no duplicate Customer_IDs were found.

- Key variables: Age; Income (annual, USD); Credit_Score (300-850); Credit_Utilization; Missed_Payments (past 12 months); Loan_Balance (USD); Debt_to_Income_Ratio; Employment_Status; Account_Tenure (years); Credit_Card_Type; Location; Month_1 to Month_6 (6-month payment history); and the target Delinquent_Account (0 = No, 1 = Yes).

- Data types: Numerical: Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure. Categorical: Customer_ID, Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6 (On-time / Late / Missed). Binary target: Delinquent_Account.

Initial data quality observations:

- Class imbalance: only 80 of 500 customers (16%) are delinquent, so accuracy alone will be a misleading metric later.

- Inconsistent labels: Employment_Status has six spellings for four categories (Employed, employed and EMP; retired in lower case).

- Scale mismatch: the guide describes Credit_Utilization and Debt_to_Income_Ratio as percentages (0-100), but they are stored as fractions (0.05-1.03 and 0.10-0.55).

- Out-of-range values: 4 customers have Credit_Utilization slightly above 100% (maximum 1.026).

Overall, the dataset is small but largely complete: 64 of 500 rows (12.8%) have at least one missing value, concentrated in Income, Loan_Balance and Credit_Score. The main quality problems are inconsistent employment labels, a few impossible utilization values, and a Missed_Payments field that does not agree with the monthly payment history. Numeric fields show no statistical outliers (IQR rule). These issues are all fixable with simple, transparent rules, and the target is imbalanced.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- **Variables with missing values:** Income (39 missing, 7.8%), Loan_Balance (29 missing, 5.8%) and Credit_Score (2 missing, 0.4%). All other columns are complete.
- **Missing data treatment:** Median imputation for all three fields, plus a "was missing" flag column for each. No columns were dropped, because none exceeds 8% missing. Synthetic data generation was considered but not used: the share missing is small, and invented values could distort the real income distribution and raise fairness concerns.

Variable	Missing	Method	Justification
Income	39 (7.8%)	Median ($107,658) + missing flag	Median is robust to extremes and income is not related to employment group in this data, so one overall median is safe.
Loan_Balance	29 (5.8%)	Median ($45,776) + missing flag	Preserves the distribution; the flag is kept because 24% of these customers are delinquent versus 16% overall (small sample, n = 29).
Credit_Score	2 (0.4%)	Median (586)	Only two records, so imputation has negligible effect and avoids losing rows.

- **Other fixes applied:** Employment_Status labels standardized to Employed, Unemployed, Retired and Self-Employed; Credit_Utilization values above 1.0 capped at 1.0 (4 records).
- **Check after treatment:** all 500 rows are now complete; means and standard deviations of the three imputed fields changed by less than 1% (for example, mean Income 108,380 before and 108,324 after).

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- Correlations observed between key variables: all numeric variables have very weak linear correlation with Delinquent_Account (|r| below 0.05, none statistically significant): Income 0.04, Account_Tenure -0.04, Credit_Score 0.04, Credit_Utilization 0.03, Debt_to_Income_Ratio 0.03, Age 0.02, Missed_Payments -0.03, Loan_Balance -0.01. Differences between segments are more informative than correlations, as shown below.

Risk indicator	Evidence in this data	Why it matters
Credit_Utilization	Delinquency rate 12.0% at 30% utilization or less, rising to 20.0% at 70-90%	Heavy use of available credit signals financial stress.
Debt_to_Income_Ratio	9.8% when DTI is 0.20 or less, versus 15-19% above 0.20	High debt relative to income reduces repayment capacity.
Employment_Status	Unemployed 19.4%, Employed and Self-Employed 16.2%, Retired 11.5%	Income stability affects the ability to pay on time.
Credit_Card_Type	Business 21.3% and Student 17.9%, versus Platinum 11.8% and Standard 12.8%	Card product reflects customer segment and limits.
Location	Los Angeles 19.6% and Houston 16.8%, versus New York 12.0%	Regional economic conditions may affect collections.
Payment history (Month_1-6)	Customers with 5 missed months: 25% delinquent (only 8 customers)	Recent behaviour is usually the strongest predictor; needs a stronger engineered feature.

Segment differences are suggestive only: chi-square tests for employment, card type and location were not significant (p = 0.55, 0.35, 0.66), so these need confirming on a larger sample.

- Unexpected anomalies:

- Missed_Payments conflicts with payment history: it matches the number of "Missed" months in Month_1-6 for only 14% of customers (correlation 0.01). The two fields are either measured over different windows or one is unreliable.

- Counter-intuitive direction: customers with more missed payments are not more delinquent (0-2 missed: about 17-18%, 3 missed: 11.5%). Month-level history also shows little separation between delinquent and non-delinquent customers.

- Credit_Score: scores are spread almost evenly from 301 to 847 and 37% are below 500, which is far lower than a typical customer base. The 500-580 band has the lowest delinquency (8.5%), which is unexpected.

- Utilization at the extreme: customers above 90% utilization (14 records) show 0% delinquency, which is counter-intuitive but based on a tiny group.

- Logical inconsistencies: 81 customers have Account_Tenure longer than their adult years (Age minus 18), and 75 have Loan_Balance above annual Income. These should be validated with Geldium.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

- 'Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.'

- 'Suggest an imputation strategy for missing values in this dataset based on industry best practices.'

- 'Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.'

- 'Check whether Missed_Payments is consistent with Month_1 to Month_6 and whether any values fall outside the ranges in the Dataset Description Guide.'

Claude (Anthropic) was used to run the analysis and draft text. Its outputs were checked against the data: missing counts, delinquency rates, correlations and chi-square tests were computed directly from the uploaded file.

6. Conclusion & Next Steps

The dataset is clean enough to model once the fixes above are applied, but the signal in the raw fields is weak. Credit utilization, debt-to-income ratio, employment status and card type show the clearest differences in delinquency rates, while payment history and Missed_Payments need further work. Data quality issues to resolve with Geldium are the Missed_Payments mismatch, the Credit_Score distribution and the scale of the ratio fields.

Recommended next steps:

1. Confirm with Geldium how Missed_Payments is defined and why it disagrees with Month_1-6.

2. Engineer features from payment history (number of late and missed months, recent trend) and combine utilization with DTI.

3. Use stratified train/test splits and evaluate with recall, precision and AUC, since only 16% of customers are delinquent.

4. Start with interpretable models (logistic regression, decision tree) before tree ensembles, and check results across employment, location and card type for fairness.

5. Request more data if possible; with 80 delinquent cases, results will be unstable.

Appendix: Dataset Description Guide (Updated)

This guide is the original Dataset Description Guide, updated to match what was actually found in Geldium's dataset (500 records, 19 columns). The last column shows the observed values and any changes made during cleaning. Three extra columns were added in the cleaned file: Income_Was_Missing, Loan_Balance_Was_Missing and Credit_Score_Was_Missing (1 = value was imputed).

Column	Description	Type	Observed in data / changes made
Customer_ID	Unique identifier for each customer.	Categorical	500 unique IDs (CUST0001-CUST0500); no duplicates.
Age	Customer age in years.	Numerical	18 to 74; no missing values.
Income	Annual income in USD.	Numerical	15,404 to 199,943; 39 missing (7.8%), filled with median 107,658.
Credit_Score	Credit score, typically 300-850.	Numerical	301 to 847; 2 missing, filled with median 586. 37% are below 500.
Credit_Utilization	Share of available credit in use.	Numerical	Stored as a fraction (0.05-1.03), not 0-100%. 4 values above 1.0 capped at 1.0.
Missed_Payments	Missed payments in the past 12 months.	Numerical	0 to 6. Matches Month_1-6 "Missed" count for only 14% of customers; flagged for review.
Delinquent_Account	Has a delinquent account (0 = No, 1 = Yes).	Binary (target)	420 No (84%), 80 Yes (16%).
Loan_Balance	Total outstanding loan balance in USD.	Numerical	612 to 99,620; 29 missing (5.8%), filled with median 45,776.
Debt_to_Income_Ratio	Total debt divided by income.	Numerical	Stored as a fraction (0.10-0.55), not 0-100%. No missing values.
Employment_Status	Current employment status.	Categorical	Six spellings cleaned to four: Employed (240), Unemployed (93), Retired (87), Self-Employed (80).
Account_Tenure	Years the customer has had an active account.	Numerical	0 to 19. 81 records have tenure longer than adult years (Age minus 18).
Credit_Card_Type	Type of credit card held.	Categorical	Gold, Student, Business, Standard, Platinum. Business and Student are not in the original guide examples.
Location	Region or city of residence.	Categorical	Five cities: Los Angeles, Phoenix, Chicago, Houston, New York.
Month_1 to Month_6	Payment history over the past 6 months.	Categorical	Stored as text (On-time, Late, Missed), not codes 0 / 1 / 2. No missing values.